Unit I. Introduction to Animation

Lesson 1 Animation History

Introduction

What is Animation?



Animation is the art of bringing static images to life by creating the illusion of movement. It involves sequencing images or frames to show gradual changes in position, shape, or appearance, simulating real motion when viewed in rapid succession. This technique can range from simple stick-figure sequences to complex, realistic character animations in movies and video games.

Animation uses a series of techniques and principles—such as **timing, spacing, squash and stretch, anticipation**, and **follow-through—** to make the movements feel natural and engaging. Animators use these principles to manipulate characters, objects, and backgrounds in ways that capture the audience's attention, tell a story, or explain complex ideas

In addition to entertainment, animation plays a crucial role in education, engineering, advertising, and medicine. In engineering, for instance, animation is a powerful tool for visualizing technical concepts and processes, which can help convey ideas more clearly to non-expert audiences.

Learning Objectives

- Understand the Evolution of Animation Techniques
- Recognize Influential Animators and Studios
- Analyze the Impact of Technological Advancements
- Relate Historical Context to Modern Animation Practices

Discussion

The History of Animation

The history of animation is a journey from rudimentary experiments with moving images to today's high-tech digital masterpieces. Here's an overview of some key developments:

Early Beginnings (Pre-1800s)

Animation's roots can be traced back to early attempts to depict motion through static images. Ancient civilizations used sequential images in pottery and tapestries to tell stories. By the late 17th and 18th centuries, inventions like the **magic lantern** (a precursor to the slide projector) were used to project images in rapid succession, giving the illusion of motion.

The 19th Century: Birth of Optical Toys

The 19th century saw the development of optical devices that paved the way for modern animation:

Thaumatrope (1824): A disk with images on both sides attached to strings. When spun, the images appeared to merge, creating a simple form of animation.

Phenakistoscope (1832): A spinning disc with a sequence of images that, when viewed in a mirror, created a looping animation.

Zoetrope (1834): A cylindrical device with slits that allowed viewers to see animated sequences as it spun.

Praxinoscope (1877): An improvement on the zoetrope, using mirrors to provide a clearer image, invented by **Émile Reynaud.**

These devices, known as "optical toys," captivated audiences by creating brief, looping animations, setting the foundation for the development of animated films.

The Early 20th Century: Birth of Film Animation

With the invention of film and camera technology, animation made significant advancements:

1908: **Émile Cohl**, a French artist, created one of the first known animated films, *Fantasmagorie*, using simple line drawings on paper. This short film is considered one of the earliest examples of traditional animation.

1914: **Winsor McCay**'s *Gertie the Dinosaur* introduced the concept of a personality-driven character in animation, capturing audiences with the lifelike motion of a dinosaur.

The Golden Age of Animation (1920s–1950s)

The 1920s to 1950s marked the golden age of animation, led by the creation of iconic animated characters and studios:

1928: **Walt Disney** introduced *Steamboat Willie*, featuring **Mickey Mouse**, one of the first

animations with synchronized sound. Disney's work became influential in animation, known for storytelling and character development.

1937: Disney released *Snow White and the Seven Dwarfs*, the first full-length animated feature film, which was groundbreaking in its use of color, sound, and character animation. This era also saw the rise of Warner Bros. and MGM, introducing iconic characters like **Bugs Bunny** and **Tom and Jerry.**

The Rise of Computer Animation (1960s–1990s)

The next wave in animation was driven by advancements in computer technology:

1960s-70s: Early computer-generated imagery (CGI) began to emerge, with simple 3D animations used in experimental films and visual effects.

1980s: Disney and other studios experimented with computer-assisted animation to enhance traditional 2D animations. The film *Tron* (1982) featured extensive use of CGI, setting a precedent for future films.

1995: Pixar's *Toy Story*, the first fully computer-animated feature film, marked a turning point in animation, showcasing the storytelling potential of 3D CGI.

Modern Animation (2000s–Present)

Today's animation combines traditional principles with digital technology, allowing for greater creativity and efficiency. Animation has expanded beyond film and television to include virtual reality, video games, and mobile apps:

- **3D Animation**: Now the standard in most animated films, 3D animation creates lifelike characters and worlds.
- **2D Digital Animation**: While 3D dominates, 2D animation persists, especially in TV shows, advertising, and educational videos. Modern tools like **Adobe Animate** streamline the production of 2D animation.
- **Motion Capture and Virtual Reality**: These technologies allow animators to capture real-life movements and create immersive experiences in VR and gaming.

Link to Video Recording

Activity

Activity Title: Building a Zoetrope Animation

Objective:

- To understand the principles of early animation through hands-on experience with Zoetrope.
- To create a simple animation sequence using sequential images.

Materials Needed:

- Cardstock or thick paper (for drawing the animation strip)
- Scissors
- Markers or colored pencils

- Ruler
- A cylindrical container or paper to create a Zoetrope drum
- Black construction paper (optional, for contrast inside the cylinder)
- Tape and glue
- A stick or pencil (to spin the Zoetrope)

Instructions:

1. **Introduction and Discussion (10 minutes)**:
    - Begin with a brief discussion on the history of Zoetrope and how it creates the illusion of movement. Explain how a sequence of images, each slightly different, can create the appearance of animation when viewed in rapid succession.
2. **Design Your Animation Sequence (20 minutes)**:
    - Draw a series of 8-12 small images on the cardstock, each showing a slight change in position to create movement (e.g., a ball bouncing, a character walking, a flower blooming).
    - Make sure each frame in the sequence progresses smoothly into the next, allowing the images to "animate" as they spin.
3. **Create the Zoetrope Drum (15 minutes)**:
    - Use a cylindrical container or roll cardstock into a cylinder.
    - Cut evenly spaced vertical slits around the top edge of the cylinder. These slits will act as viewing windows, allowing viewers to see only one frame at a time.
    - Optionally, line the inside of the cylinder with black construction paper for a clear contrast.
4. **Attach the Animation Strip (10 minutes)**:
    - Tape or glue your animation strip inside the cylinder so that each image aligns with a viewing slit.
5. **Test the Zoetrope (5 minutes)**:
    - Spin the Zoetrope gently while looking through the slits. Adjust the speed until you see the animation come to life!
6. **Reflection and Discussion (10 minutes)**:
    - Discuss as a class: How does seeing your animation in the Zoetrope differ from simply looking at each frame? What principles of animation (like timing and spacing) can you observe in your Zoetrope animation?

Rubrics of the Activity

Rubric for Zoetrope Activity

Criteria	Excellent (4 pts)	Good (3 pts)	Satisfactory (2 pts)	Needs Improvement (1 pt)	Points
Creativity in Animation	Unique and creative animation, smooth progression	Clear and engaging animation sequence	Animation sequence has some creativity but may lack fluidity	Animation lacks creativity or coherence	/4
Accuracy of Sequence	Each frame follows the previous one perfectly	Minor gaps in frame progression	Noticeable gaps in progression	Frames do not progress logically	/4
Craftsmanship	Well-constructed Zoetrope, sturdy and functional	Minor construction flaws, functions well	Some issues with construction, affects function	Poor construction, Zoetrope not functional	/4

Criteria	Excellent (4 pts)	Good (3 pts)	Satisfactory (2 pts)	Needs Improvement (1 pt)	Points
Understanding of Concept	Shows excellent understanding of Zoetrope concept	Shows good understanding of concept	Basic understanding of Zoetrope concept	Lacks understanding of Zoetrope concept	/4
Effort and Engagement	Demonstrated high effort, fully engaged in activity	Actively participated, completed task on time	Somewhat engaged, moderate effort	Little engagement, minimal effort	/4

Value Integration

A Brief History of Animation – Will Seymour Murphy, 2015

How to make Zoetrope – Salih Art and Tech, 2021